MANAGEMENT SUMMARY

Topic 1: Language Identification

Group 17
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OVERVIEW

Goal: To build a language identification system capable of detecting the language of text across 10 languages: Italian (it), Tamil (ta), Belarusian (be), Russian (ru), German (de), Spanish (es), English (en), Portuguese (pt), French (fr), and Korean (ko).
To do it, we implemented and compared several approaches:

- Baselines: Rule-Based system, Machine Learning Based system (Naive Bayes)
- Final Solution: Rule-Based augmented XLM-RoBERTa

MAIN CHALLENGES

- *Similar languages*: Some languages are very similar to each other (for example, Russian and Belarusian, or Romance languages), which makes them difficult to distinguish automatically.
- *Uneven data availability*: Some languages had much more training data than others, which affected model performance.
- *Large data volume*: The datasets were very large, which limited the choice of models due to time and hardware constraints.
- *Noisy real-world text*: Texts often contain spelling mistakes, slang, or mixed languages, which reduces accuracy.

EXTERNAL RESOURCES

Datasets:

- Multilingual Wikipedia dataset on Hugging Face – main training and evaluation data, clean and well-structured
  https://huggingface.co/datasets/wikimedia/wikipedia
- Twitter / social media datasets on Hugging Face and Kaggle – used for testing and evaluation on real-world, noisy text
  https://huggingface.co/datasets/pere/italian_tweets_500k
  https://huggingface.co/datasets/ehznapsxm/kor_sns_korea_5
  https://www.kaggle.com/datasets/kracekumar/tamil-binary-classification-1k-tweets-labels-v1
  https://huggingface.co/datasets/FrancophonIA/french_tweets
  https://huggingface.co/datasets/fpaulino/portuguese-tweets
  https://huggingface.co/datasets/maaxap/BelarusianGLUE
  https://huggingface.co/datasets/MonoHime/ru_sentiment_dataset
  https://huggingface.co/datasets/Alienmaster/german_politicians_twitter_sentiment
  https://www.kaggle.com/datasets/mmmarchetti/tweets-dataset/code

Libraries and Models:

- Stanza: A Python Natural Language Processing Toolkit for Many Human Languages
- XLM-RoBERTa: A pretrained multilingual language model

IMPLEMENTED SOLUTIONS

Rule-Based Baseline

- Language identification based on basic rules - writing system (e.g. Latin, Cyrillic), common words, frequent character n-grams and typical character patterns.
- Easy to understand, but struggled with similar languages.
  e.g.: RU: Мы живём и работаем тут.
  BE: Мы жывём і працуем тут.

Machine Learning Baseline (Naive Bayes)

- Trained on text features.
- Fast and simple, but less robust to noisy, short, informal text.

Final Model (Rule-Based Augmented XLM-RoBERTa)

- Finetune XLM-RoBERTa by using rule-based output as special prefix tokens <SCRIPT=CYRILLIC> <RB=ru> <RB_CONF=HIGH>
- Prepend special tokens to the input text, then the model updates weights based on new inputs.
- Overcome a rule-based model's confusion between languages that share the same script by using a deep learning approach that leverages language-specific features.
- Overcome existing deep learning models' confusion between languages with different scripts caused by vectorization using special tokens.

RESULTS

Table1. Wikipedia Dataset Results

Model	Accuracy	Precision	Recall	F1-score
Fine-tuned XLM-RoBERTa	0.9842	0.98	0.98	0.98
Rule-based	0.9488	0.95	0.95	0.95
Naive Bayes	0.8599	0.91	0.86	0.87

Table2. Twitter Dataset Results

Model	Accuracy	Precision	Recall	F1-Score
Fine-tuned XLM-RoBERTa	0.8878	0.90	0.89	0.89
Rule-based	0.8122	0.76	0.74	0.74
Naive Bayes	0.6764	0.83	0.68	0.68

- The final model achieves the best performance on both clean and noisy datasets.
- The final model is the most generalizable and performs consistently across diverse datasets.
- Despite strong overall results, performance decreases on noisy data compared to clean datasets.
- Similar languages with the same script (especially within Latin-script languages) are the most challenging cases.

POTENTIAL NEXT STEPS

- Increase training data for languages that share the same script.
- Train on noisier real-world text to build a model that is more robust to noise.
- Add more languages or improve coverage for low-resource ones.
- Optimize for deployment (smaller model or faster inference).

CONTRIBUTIONS

Olesia Galynskaia: Preprocessing, Rule-Based Model
Julia Chalissery: Preprocessing, Machine Learning Model
Yeongshin Park: Preprocessing, Final Model, Evaluation
Rebecca Micol: Preprocessing, Final Model, Evaluation